



**Tribhuvan University**  
**Faculty of Humanities and Social Sciences**

**A Project Proposal on**

**BooksMandu**

**Submitted to**  
**BCA department,**  
**Nepal Kasthamandap College**

***In partial fulfilment of the requirements for the Bachelors in Computer  
Application***

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# **CHAPTER 1: INTRODUCTION**

BooksMandu is a web based application where user can buy, sell, rent and exchange books without involvement of third party where user will get book recommendation according to their behavior. Here user can see all the details about the book like name, author, location, price condition of the book and many more without logging in but if you want to exchange your book with another book, you have to login to our system and you can sent exchange request. Based on your behavior and choice of book you are searching our system will recommend similar books next time you visit our website.

## **1.1 Problem statement:**

- a. User Preference Prediction
- b. Book Similarity
- c. Scalability
- d. Integration of External Data
- e. Personalization

## **1.2 Objectives:**

- a. To create a platform where user can buy, sell, rent and exchange book
- b. To recommend books to the user based on personalized data if user have account in our system, generalized recommendation if user is new to the system.

## **CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW**

### **2.1 Background study:**

Many existing systems in Nepal sell secondhand books which are dedicated for course books for students but there is no specific used bookstore for other genres. There are no systems where the owners and sellers both get to interact and buy, sell and exchange their used books. All systems were built as there is the owner of the books and many buyers.

There are no any recommendation system where the system can identify the behavior of the user or they cannot study the behavior of the similar user so that the system can recommend related and interesting books to the user.

### **2.2 Literature review**

#### **Trust and Perceived Usefulness**

[1] Their research underscores the importance of trust and perceived usefulness in the adoption of online platforms. For your application, ensuring a secure and reliable system where users can trust the transactions is vital. Implementing clear policies and secure payment gateways can help build this trust.

#### **Consumer Decision Factors**

[2] They explore factors influencing consumer decisions in online book purchases. Key factors include website usability, product variety, and competitive pricing. For your application, a user-friendly interface and a wide selection of books will be crucial to attract and retain users.

#### **Recommendation Systems**

[3] They provide an overview of content-based recommendation techniques. These techniques can be highly effective in personalizing user experience by suggesting books based on user behavior and preferences. Implementing an intelligent recommendation system will enhance user engagement and satisfaction.

#### **Direct Peer-to-Peer Transactions**

[4] Their study on peer-to-peer platforms highlights the importance of user reviews and ratings in building trust. Encouraging users to leave feedback and ratings can help maintain a high level of trust and transparency within the community.

#### **Economic Implications**

[5]Bakos discusses the reduction of transaction costs through electronic marketplaces. Your application can benefit users by eliminating intermediaries, thus lowering costs and increasing the efficiency of transactions.

### **User Experience Design**

[6]Nielsen's principles of usability are fundamental for creating an intuitive and user-friendly interface. By adhering to these principles, your application can ensure that users can easily navigate the platform, perform transactions, and find the books they need without frustration.

### **Social Influence and Network Effects**

[7]Their study on network effects explains how the value of a product or service increases as more people use it. For your application, leveraging social influence and encouraging user participation through features like social sharing and community discussions can create a more engaging and valuable platform.

## **CHAPTER 3: METHODOLOGY**

### **3.1 Requirement identification**

#### **3.1.1 Requirement analysis**

##### **Function requirement**

##### **User Registration and Authentication:**

In every system user registration and authentication is a mandatory and required function which deals with the security of the system.

##### **User Profile Management:**

Here users should be able to create and update their profiles, including personal information and payment methods.

##### **Book Management**

The system should allow admins and sellers to add, update, and delete book listings. Each book listing must include details such as title, author, genre, price, and a brief description.

##### **Book Recommendation System:**

Developing a recommendation system that suggests books based on user preferences, search history, and purchase history.

##### **Order Management**

Users should receive order confirmation and tracking information via email and in their user profile.

### **3.2 Feasibility study**

#### **3.2.1 Technical**

There are multiple tools and techniques are present in the market that support recommendation system so it is technically feasible.

#### **3.2.2 Operational**

We can continuously get data from the user and process it from in the engine and recommend user accordingly so it is operationally feasible.

#### **3.2.3 Economical**

Once we create this system we only need to repair it and system will feed the data and only database is needed to store the data and timely enhance the algorithm so this system is economically feasible as well.



### 3.3 High level design of system

#### 3.3.1 Methodology and working mechanism of the proposed system

Here in the methodology I will use agile methodology because it gives us freedom to change our system as per requirement since I don't have clear requirement for the system.

Here in the working mechanism it will perform the following function.

##### 1. User interaction:

Here from user interaction the system will deal with the user and interact with the user. System will also study the behavior of the user and store the data for recommendation system.

##### 2. Book category:

Here in book category, we divide the books into multiple parts.

##### 3. Searching and browsing:

Here, the system will help the user to search the book according to its book name, author name, location, genres and all the data will be stored by the recommendation system.

##### 4. Recommendation system:

Here, all the data gathered about the user will be analyze and recommend the user according to the analysis.

#### 3.3.2 Flow chart

[8]A flowchart is a picture of the separate steps of a process in sequential order. Here is a flow chart of BooksMandu.

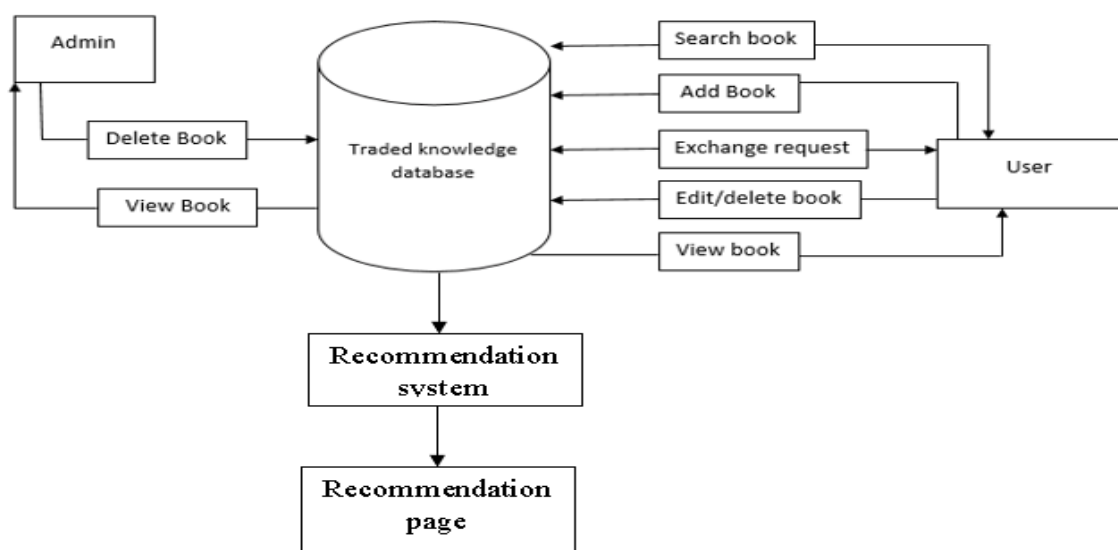


Figure 1: Flow chart

### 3.3.3 Data modeling ER:

[9]An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research. Here is a er diagram for BooksMandu.

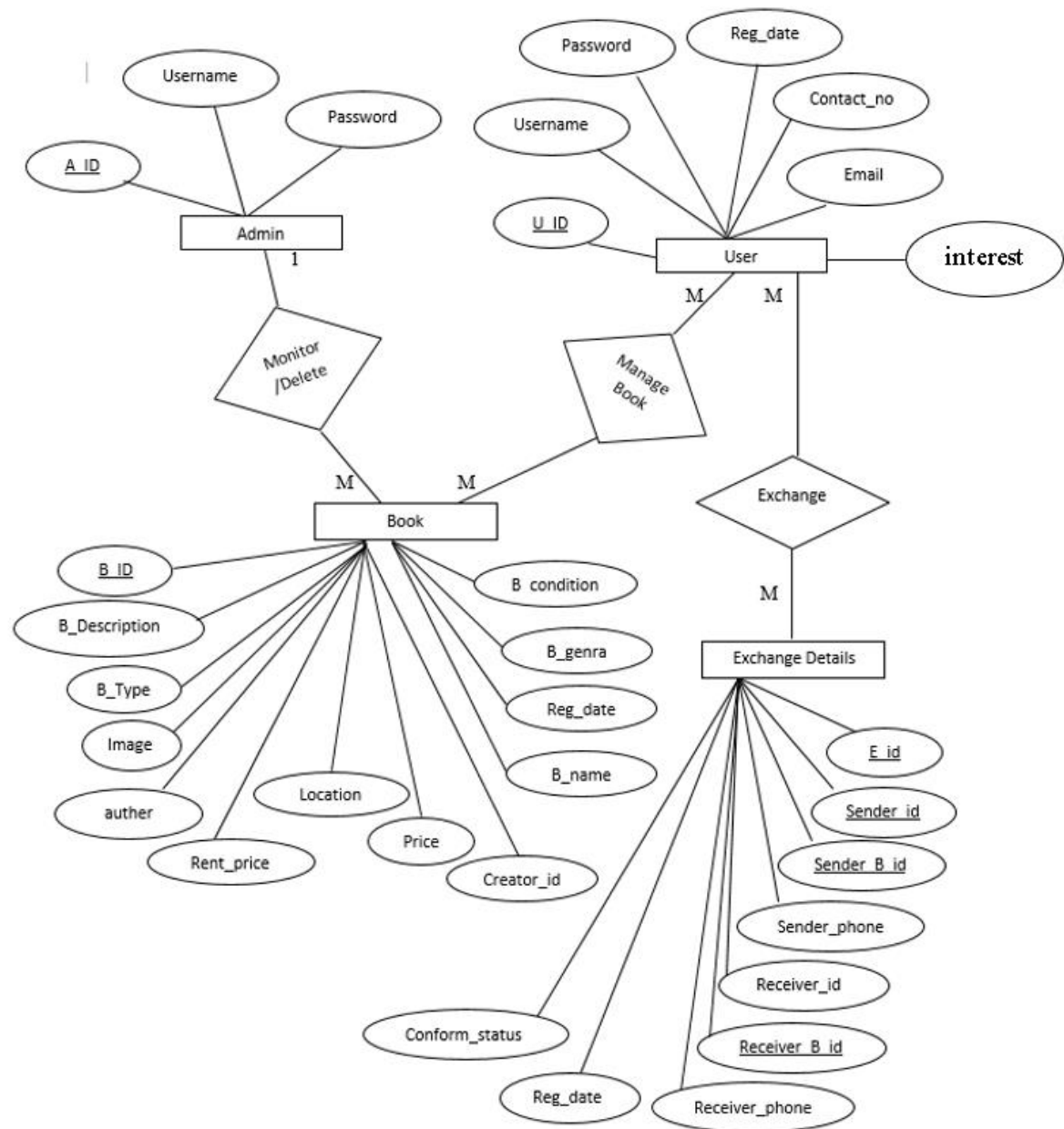
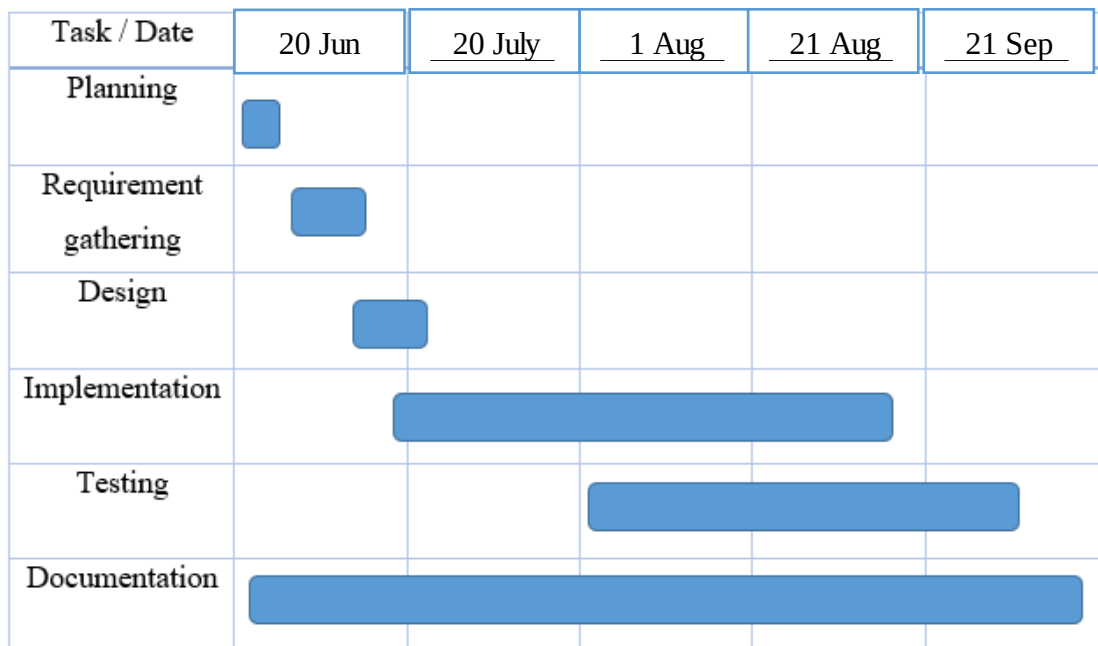


Figure 2: ER diagram

### 3.4 Gantt chart

[10]A Gantt chart is a project management tool assisting in the planning and scheduling of projects of all sizes; they are particularly useful for visualising projects. A Gantt chart is defined as a graphical representation of activity against time; it helps project professionals monitor progress.

**Table 1: Gantt chart**



## **CHAPTER 4: EXPECTED OUTCOMES**

Here are the list of expected outcomes of BooksMandu.

1. Web based application where user can buy, sell, rent and exchange book online without involvement of third party.
2. Book recommendation based on general user if user is new, personalized if the user have account in our system.

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